

Introduction to Computational Biomedicine and Health

BINF GU4001/COMS W4560

Mondays and Wednesdays, 4:10-5:25PM

Location: Mudd 327 ([Map](#))

Columbia University
Fall 2026

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Course Description

This course offers a comprehensive introduction to the core computational methods driving modern biomedical research and health data science. As biological and clinical datasets grow in scale and complexity from genomic sequences and molecular profiles to electronic health records (EHRs) and consumer health data this course equips students with the essential computational foundations to model, analyze, and interpret high-dimensional biomedical data.

Incorporating key algorithmic challenges spanning Bioinformatics, Clinical Informatics, Consumer Health Informatics, and Public Health Informatics, the course focuses on the design and application of algorithms and statistical models to solve real-world biomedical problems. Lectures emphasize practical techniques and showcase their use across diverse biomedical data types.

Designed for advanced undergraduates and graduate students in biomedical informatics, computer science, biomedical engineering, applied mathematics, and related fields, this course builds a rigorous understanding of computational biomedicine. It is cross-listed with Computer Science and serves as a core requirement for the Biomedical Informatics PhD and master's programs as well as the Health and Medicine concentration of the Master of Science in Artificial Intelligence program. Students interested in careers in biomedical informatics, health data science, computational biology, or biomedical AI will find this course especially valuable.

As biomedical data continue to expand in size, diversity, and impact, this course provides a critical foundation in the algorithmic, statistical, and computational tools needed to advance the future of research at the intersection of computation and human health.

Learning Objectives

By the end of the course, students will be able to:

- Understand the core computational methods that underpin biomedical research and health data science, including algorithm design, statistical modeling, and data integration techniques applied to biological and clinical data.
- Identify and analyze key challenges in Bioinformatics, Clinical Informatics, Consumer Health Informatics, and Public Health Informatics and learn how to apply appropriate computational approaches to address real-world problems in each domain.
- Develop proficiency in understanding computational studies using high-dimensional data such as genomic sequences, molecular phenotypes, electronic health records (EHRs), and wearable sensor data, with an emphasis on application of informatics and AI.
- Critically evaluate biomedical studies and data-driven health applications, including understanding issues of bias, generalizability, reproducibility, and ethical considerations in biomedical data science.

These learning goals are designed to give students a rigorous foundation in computational biomedicine while fostering problem-solving, critical thinking, and communication skills essential for research and careers in health data science, biomedical informatics, and biomedical AI.

Prerequisites

There are no formal prerequisites. While prior experience with informatics is helpful, it is not required.

Required Texts and Materials: There are no required course textbooks. All required lecture slides, readings, and supplemental materials will be provided through CourseWorks or linked from the course website ahead of the lectures.

Course policies

Communication: Announcements will be posted on CourseWorks. Regularly check for updates and use the platform for submitting assignments and asking questions.

Attendance and Participation: This is an in-person course, and regular attendance is expected because lectures, discussions, and in-class activities contribute directly to the learning objectives. Students may miss up to three regular class meetings without concern. Participation grades are based on substantive engagement with course discussions and activities rather than attendance alone. Students who anticipate absences covered by University policy should contact the Course Director so that appropriate arrangements can be made. Students will not be penalized for absences due to religious observance and will be provided an equivalent opportunity to satisfy missed academic requirements. Please notify the Course Director as early in the semester as possible.

Academic Accommodations: Students who require disability-related academic accommodations should register with Columbia Disability Services (DS) and ensure that the instructional team receives the appropriate accommodation notification. Approved accommodations will be implemented in accordance with University policy and handled confidentially. Students who believe they may require an accommodation are encouraged to contact Disability Services as early as possible.

Academic Integrity: Students are expected to submit their own work and to comply with Columbia University's standards of academic integrity, including the Faculty Statement on Academic Integrity and applicable undergraduate and graduate academic integrity policies. Plagiarism, unauthorized collaboration or assistance, fabrication, and other forms of academic dishonesty are prohibited. Suspected violations may result in an academic sanction and referral to the appropriate University academic integrity or Dean's Discipline process. Unless explicitly authorized for a particular assignment, the use of generative AI tools (including for generating, rewriting, summarizing, or substantially editing material submitted for credit) is prohibited. Students are responsible for asking the instructional team if they are uncertain whether a particular use is permitted.

Course Requirements and Grading Breakdown

Students will be evaluated based on multiple assignments designed to assess their understanding and application of core concepts in computational biomedicine and health. Grading is based on academic performance with clear criteria provided for each component.

Grading Breakdown

Component	Percentage of Final Grade	Details
Class Participation	10%	Active participation is expected.
Lecture Summaries	20%	Due by the beginning of the next class meeting.
Group Project	30%	Final slides must be uploaded to CourseWorks by Monday, December 7, 2026, 4:00 PM. Presentations will occur during the last two class sessions; all students must attend both sessions.
Term Paper	40%	Due December 17, 2026, at 11:59 PM ET.
Total	100%	

Based on points earned for the lecture summaries, term paper, group project and class participation, the class grade will be assigned according to the following scale:

A	94-100%	B+	87-89%	C+	77-79%	D	60-69%	F	<60%
A-	90-93%	B	83-86%	C	73-76%				
		B-	80-82%	C-	70-72%				

Rubrics and Evaluation Criteria

Lecture Reflection Summaries (20% of Final Grade) | Each summary is due by the next class lecture

Following each lecture, students are required to submit a concise summary of the material covered (500 words maximum, 12pt Times New Roman font, single-spaced), focusing on core concepts and key takeaways. Students should *capture*, *connect*, and *question*: What were the core concepts and examples? How does this lecture relate to other domains or methods? What remains unclear, contested, or worth exploring?

Summaries are due by the start of the next class session and must be submitted via CourseWorks. Each homework is worth 1 point, and students are allowed to miss up to 3 submissions without penalty. The overall score will be normalized to 20% of the final course grade.

Criteria	Excellent (1 point)	Incomplete (0 points)
Content Understanding	Clearly and accurately captures key topics, concepts, and discussions from the lecture.	Summary is missing, incomplete, or does not reflect key content.
Clarity and Writing	Well-organized, coherent, and clearly written.	Poorly organized or difficult to follow.
Originality	Written in the student's own words without copying slides or external materials.	Largely copied or generated by unauthorized tools (including AI).

Term Paper (40% of Final Grade)

Proposal due November 2, 2026, at 11:59 PM ET; Final paper due December 17, 2026, at 11:59 PM ET.

A written research paper exploring a topic in biomedical informatics. The paper should demonstrate deep engagement with course concepts and independent research. This paper enables students to delve more deeply into a topic of personal interest in the context of biomedical informatics. Generally, this paper should be either a review of a particular issue or technology in biomedical informatics, or a well-researched position paper expressing a personal view on a controversial topic in the field.

The paper must have labeled sections for Background, Current Approaches, Discussion, and Conclusions. Grading will be based primarily on the writer's ability to discuss a compelling argument about the chosen topic, with at least three concrete examples, identified gaps in research, and citations. Sources should consist primarily of peer-reviewed scientific literature; authoritative technical standards, government publications, and professional guidelines may also be cited when appropriate. The discussion section focuses on limitations from the current approaches and possible directions for future work for the selected topic within the scope of feasibility. The paper must include your name and a title, and be a maximum of 2500 words (not including figures and references), double spaced, 12-point Times New Roman font. Any citation style is acceptable as long as it is consistent throughout the essay. Spelling, grammar, and overall readability will be considered.

Criteria	Excellent (A-level)	Good (B-level)	Needs Improvement (C or below)
Topic and Scope	Focused, original, and clearly relevant to informatics.	Relevant topic but somewhat broad or underdeveloped.	Topic lacks relevance or clarity.
Research and Evidence	Strong use of scholarly sources, well-integrated evidence.	Adequate sources with some gaps or weak integration.	Insufficient or inappropriate sources.
Insight	Demonstrates critical thinking and original insights.	Some insight but primarily descriptive.	Little to no insight; mostly summary.
Organization and Clarity	Logical structure, clear argument, and smooth flow.	Some structural issues or unclear sections.	Disorganized and difficult to follow.
Writing Quality	Grammatically correct, polished, and professional.	Minor grammatical errors.	Frequent grammatical issues.
Citation	Sources are appropriately and consistently cited throughout.	Minor citation or formatting inconsistencies.	Sources are inadequately or inconsistently cited.

Proposal Instructions: Students must submit a one-paragraph proposal through CourseWorks describing their proposed term paper topic, the central question or argument they plan to address, and its relevance to biomedical informatics. The proposal is intended to ensure that the topic is appropriate in scope and aligned with the expectations of the assignment.

Group Project Presentation – 30% of Final Grade

The instruction team will assign students into **groups of 3-4**, which will be announced on **October 5, 2026**. Students will work collaboratively within their group to design and present a **computational solution** addressing a biomedical problem of their choice. Each project must explicitly incorporate at least two of the following domains: bioinformatics, clinical informatics, consumer health informatics, and public health informatics. The presentation will take place during the last two sessions of class.

All groups must submit their presentation slides (**PPT, PPTX, or PDF format**) to **CourseWorks** by **Monday, December 7, 2026, 4:00 PM**. Attendance is required during all final presentations, except for absences covered by University policy.

Required Presentation Content: Each section below should have at least one dedicated slide. Additional slides are encouraged for clarity and depth.

1. **Modules:** Explicitly state the course domains integrated (bioinformatics, clinical informatics, consumer health informatics, public health informatics).
2. **Problem:** Clearly define the biomedical problem or question. Specify if focusing on a disease, behavior, or another biomedical issue.
3. **Current Approaches:** Describe existing methods addressing the problem and clearly explain your project's novel contribution or improvement.
4. **Population:** Clearly specify who benefits from your solution (e.g., researchers, clinicians, patients).
5. **Data:** Describe available data sources, their relevance, and how they can be integrated effectively.
6. **Methods:** Clearly outline the computational solution, detailing the methodological approach and execution plan.
7. **Evaluation:** Define clear, specific criteria for assessing your solution's success, including potential outcomes and validation strategies.
8. **Delivery Method:** Clarify your deliverable, the intended audience, and how it will be provided or implemented.

Presentation Evaluation Criteria:

Criterion	Excellent	Good	Needs Improvement
Content	Comprehensive and clearly covers all required sections with strong depth and accuracy.	Covers required sections but with limited depth or minor inaccuracies.	Sections incomplete, superficial, or unclear.
Delivery	Clear, confident, engaging, and professional.	Adequate delivery, some hesitation or minor engagement issues.	Delivery unclear, disengaged, or lacking professionalism.
Visual Aids	High-quality visuals that significantly enhance understanding and clarity.	Adequate visuals, minor improvements needed.	Visuals poor quality, unclear, or absent.
Timing and Flow	Presentation well-paced, logically structured, within allotted time.	Slight timing issues, minor structural concerns.	Poor pacing, significantly over/under time, disorganized flow.

Course Schedule

Lecture	Date	Title
1	Wednesday, 09/09/26	Introductory Lecture
2	Monday, 09/14/26	Human-Centered AI
3	Wednesday, 09/16/26	Introduction to Machine Learning
4	Monday, 09/21/26	Health Data Ecosystems and Standards
	Wednesday, 09/23/26	No Class
5	Monday, 09/28/26	Clinical Decision Support
6	Wednesday, 09/30/26	Introduction to Bioinformatics & Genetics
7	Monday, 10/05/26	Generating Real-World Evidence
8	Wednesday, 10/07/26	History of AI in Medicine
9	Monday, 10/12/26	Representation Learning with Health Data
10	Wednesday, 10/14/26	Cognitive Informatics
11	Monday, 10/19/26	Clinical Research Informatics
12	Wednesday, 10/21/26	Human-Computer Interaction
13	Monday, 10/26/26	Phenotyping
14	Wednesday, 10/28/26	Systems Biology
	Monday, 11/02/26	Academic Holiday – No Class
15	Wednesday, 11/04/26	Biomedical Privacy
16	Monday, 11/09/26	Term Paper Q&A and Group Project Meetings
17	Wednesday, 11/11/26	Public Health Informatics
18	Monday, 11/16/26	Introduction to Microbiome Data Analysis
19	Wednesday, 11/18/26	Computational Genetics
20	Monday, 11/23/26	Biological Foundation Models
	Wednesday, 11/25/26	Academic Holiday – No Class
21	Monday, 11/30/26	Consumer Health Informatics
22	Wednesday, 12/02/26	Personal Health Agents
23	Monday, 12/07/26	AI Implementation and Applications
24	Wednesday, 12/09/26	Student Presentations
25	Monday, 12/14/26	Student Presentations